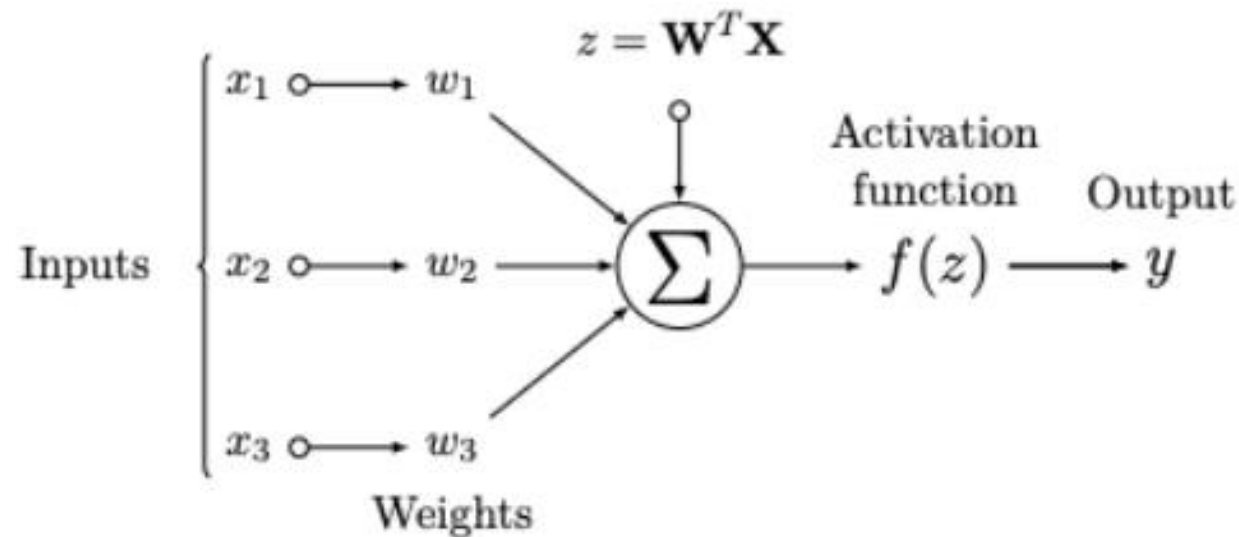


ACTIVATION FUNCTION

- An Activation Function decides whether a neuron should be activated or not.
- It decide whether the neuron's input to the network is important or not in the process of prediction using simpler mathematical operations.



Activation Function Types:

It uses threshold to produce an output

- Unit or Binary
- Sigmoid or Logistic
- TanH
- ReLU
- Softmax

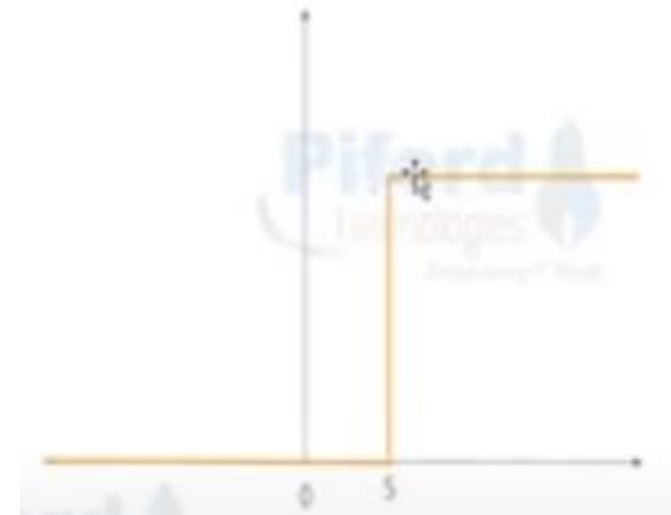
1 - Unit or Binary:

The output of unit step function is 1 or 0.

$$f(x) = \begin{cases} 0 & \text{if } \theta > x \\ 1 & \text{if } x \leq \theta \end{cases}$$

Threshold value is 5

x	f(x) = x	f(x)
4	f(4)=0	0
6	f(6)=1	1
1	f(1)=0	0
8	f(8)=1	1

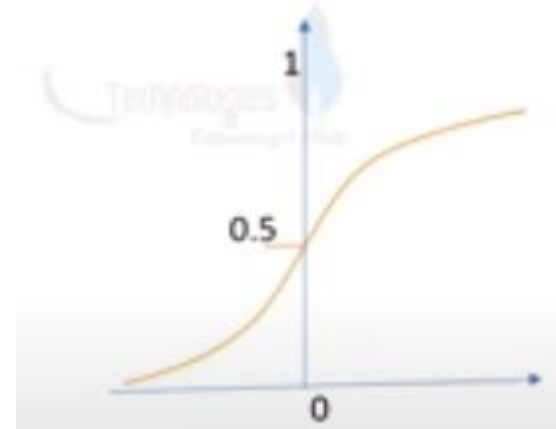


2- Sigmoid or Logistic

When to use Sigmoid function?

Ans. When we want to map input values to a value in the range of 0 to 1, we apply sigmoid transformation function.

$$f(x) = \frac{1}{1 + e^{-x}}$$



Input can span between – infinity to + infinity

x	f(x) = x	f(x)
1	f(1) = 0.73	0.73
10	f(10) = 0.99	0.99
100	f(100) = 1	1
-1	f(-1) = 0.268	0.268
-10	f(-10) = 4.539	4.539
-100	f(-100) = 3.720	3.720

```
In [9]: import math

def sigmoid(x):
    return 1 / (1 + math.exp(-x))

sigmoid(1)
```

Out[9]: 0.7310585786300049

```
In [11]: sigmoid(10)
```

Out[11]: 0.9999546021312976

```
In [13]: sigmoid(100)
```

Out[13]: 1.0

```
In [15]: sigmoid(-1)
```

Out[15]: 0.2689414213699951

```
In [17]: sigmoid(-10)
```

Out[17]: 4.5397868702434395e-05

```
In [18]: sigmoid(-100)
```

Out[18]: 3.7200759760208356e-44

3- Tanh

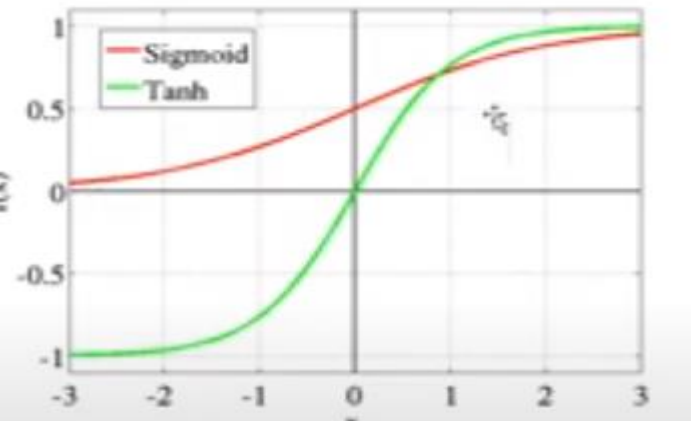
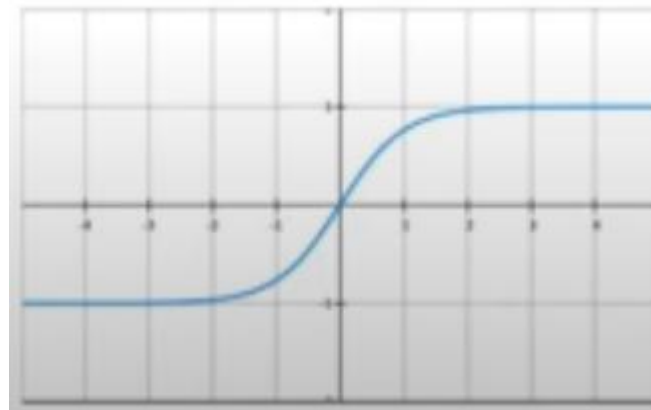
Tanh(x) gives the hyperbolic tangent of x

$$f(x) = \tanh(x) = \frac{2}{1 + e^{-2x}} - 1$$

Unlike Sigmoid function, the range of Tanh is -1 to +1 . The advantage of tanH is that we can deal negative numbers.

$$f(x) = \tanh(x)$$

x	f(x) = x	f(x)
-4	f(-4) = -1	-1
-2	f(-2) = -0.7	-0.7
2	f(2) = 0.7	0.7
4	f(4) = 1	1



Sigmoid work for positive values.

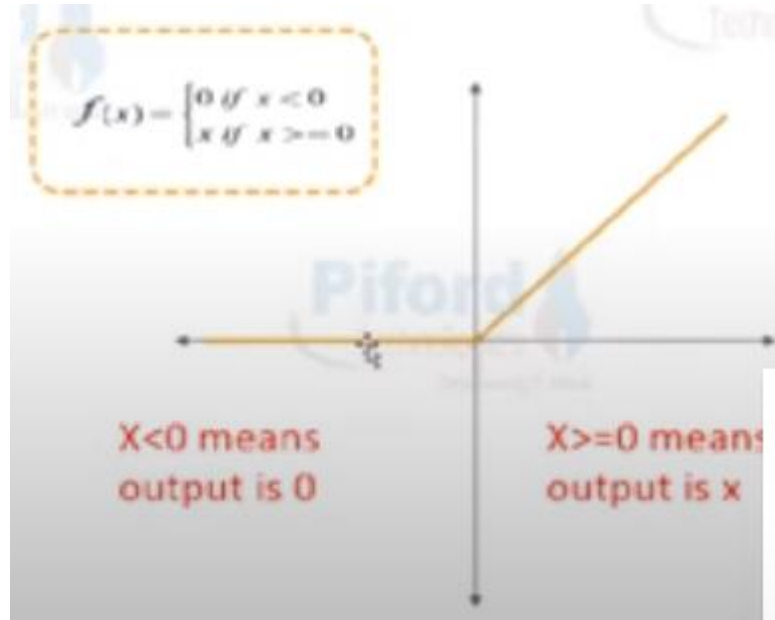
TanH work for positive and negative values.

4- ReLU

If input is less than 0 then output would be 0 but if the input is greater than threshold value then output would be x

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

x	f(x) = x	f(x)
-3	f(-3) = 0	0
-5	f(-5) = 0	0
3	f(3) = 3	3
5	f(5) = 5	5



5- Softmax

It is used in classification problems when we have multiple classes. It will tell us Probability of each class and we can then find the class with the highest probability.

X	Class A	Class B	Class C	Class D
1	0.97	0.1	0.1	0.1
11	0.3	0.3	0.91	0.3
23	0.2	0.94	0.2	0.2
9	0.3	0.1	0.5	0.1